

Hub Gene Annotation: Automated Pipeline and Manual Analysis

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1 Overview

After running signed WGCNA on the TCGA-BRCA tumor expression matrix, we obtained nine co-expression modules and extracted the top 20 hub genes per module by absolute module membership (kME), giving 180 module-gene entries that correspond to 165 unique genes. The script `annotate_hub_genes.py` automates the part of that work that can be automated, and produces a single consolidated table. This document explains exactly what the script does, what each database is, and what still has to be done manually.

2 Background

Different databases name the same gene in different ways, and the script has to translate between them. Three identifier systems matter here.

- **Ensembl gene ID** (for example `ENSG000000091831`). This is the identifier used inside our expression matrix and in the hub gene file. Ensembl IDs often carry a version suffix such as `.24`. The version suffix is not part of the real identifier, so the script removes it before any lookup. Leaving it on causes silent lookup failures.
- **HGNC symbol** (for example `ESR1`). This is the human readable gene name. It is convenient but unstable, because one symbol can be an alias for more than one gene, which is why the script does not query by symbol first.
- **Entrez (NCBI) gene ID** (for example `2099`). This is a plain integer used by NCBI and by cBioPortal. The script obtains it from the first database it queries and then reuses it to talk to cBioPortal.

The practical consequence is that the script uses the stable Ensembl ID as its primary key, falls back to the symbol only when an Ensembl ID does not resolve, and carries the Entrez ID forward for the cancer database step.

3 The databases the script uses

3.1 MyGene.info (gene biology)

MyGene.info is a free programming interface. It does not generate its own biology; it aggregates and serves content that originates at NCBI, Ensembl, and RefSeq. For each gene it can return the official full name, a written functional summary, a list of aliases, and the Entrez ID. The summary text it returns is the curated NCBI RefSeq gene summary, which is the same paragraph on the NCBI Gene website. The advantage of the API is that we can request all 165 genes in a single batch call instead of opening 165 web pages.

3.2 cBioPortal and TCGA

The Cancer Genome Atlas (TCGA) is a large public project that sequenced thousands of tumors, including a breast cancer cohort (BRCA). cBioPortal is a portal that hosts TCGA and many other cancer studies and, importantly, exposes a documented API so the data can be queried by a script. The study the script targets by default is the curated breast cancer PanCancer Atlas cohort.

For each hub gene the script asks cBioPortal two questions about that cohort:

- **Mutation frequency.** In what fraction of sequenced tumors does this gene carry a mutation?
- **Copy number alteration frequency.** In what fraction of tumors is the gene deeply deleted or amplified? Copy number refers to how many copies of the gene the tumor has. The script counts only the strong events, namely deep deletions and amplifications, which cBioPortal derives with an algorithm called **GISTIC**. Mild single copy gains and losses are not counted, because they are common background noise.
- **NOTE:** *We do not set the thresholds for the 'deep' deletion or amplification, it is inbuilt in the GISTIC algorithm.*

It then reports a combined `Altered_pct`, which is the fraction of profiled tumors that have either kind of event. A high altered percentage is a signal that the gene is not just a co-expression hub in our network but is also genetically disturbed in real patients, which makes it a stronger candidate for follow up.

4 How the code works

1. **Load the hub gene CSV.** It reads the file with columns `Module`, `EnsemblID`, `GeneSymbol`, and `kME`, and checks that all four are present.
2. **Clean the Ensembl IDs.** It strips the version suffix so that, for example, `ENSG00000091831.24` becomes `ENSG00000091831`.
3. **Deduplicate.** The 180 rows reduce to 165 unique genes for querying, so a gene that is a hub in two modules is looked up only once and then mapped back onto both of its rows.
4. **Query MyGene.info.** It sends the 165 Ensembl IDs in one batch request and receives names, summaries, aliases, and Entrez IDs. Any gene that does not resolve by Ensembl ID is retried using its symbol.
5. **Query cBioPortal.** It collects the Entrez IDs gathered in the previous step and makes two batch requests against the breast cancer cohort, one for mutations and one for copy number, then computes the three percentages per gene. This step is wrapped so that if cBioPortal is unreachable or an endpoint name has changed, the rest of the table is still written, with the three cancer columns left blank.
6. **Write the consolidated CSV.** Every original row is preserved and the new annotation columns are attached.

4.1 Columns in the output table

Column	Meaning
Module	WGCNA module the gene is a hub of
GeneSymbol	Human readable gene name
EnsemblID	Original Ensembl ID with version
kME	Absolute module membership, how central the gene is to its module
Full_Name	Official gene name from NCBI
Aliases	Other names the gene is known by
EntrezID	NCBI integer ID, used for the cancer lookup
BRCA_Mutation_pct	Percent of TCGA-BRCA tumors with a mutation in the gene
BRCA_CNA_pct	Percent with a deep deletion or amplification
BRCA_Altered_pct	Percent with either event
Functional_Summary	NCBI RefSeq description of what the gene does

5 Remaining Analyses

5.1 Condensing the GO enrichment dotplots

Each module produced GO enrichment results across three branches: Biological Process, Cellular Component, and Molecular Function. A dotplot shows the most enriched terms, where the position reflects how strongly the term is overrepresented and the dot size reflects how many of our genes fall into it.

- **Reducing redundancy.** GO contains many near duplicate parent and child terms (for example cell cycle, cell cycle process, mitotic cell cycle). Paste the top enriched terms for a module into REVIGO, or the `rrvgo` R package, to collapse them into a few representative concepts. This turns a crowded plot into a short list.
- **Name each module by function.** Replace the color label with a biological label.
- **Two more resources.** Whereas GO only says something broad like metabolic process, we can run the module hub genes through:
 - **MSigDB** (the Molecular Signatures Database), specifically its Hallmark gene sets and oncogenic signature sets, which report directly whether our genes match defined cancer programs.
 - **ConsensusPathDB**, which aggregates KEGG, Reactome, and BioCarta pathways at once and will often replace a vague GO term with a precise pathway name.

5.2 Clinical validation of the selected hub genes

The consolidated CSV is meant to narrow 165 genes down to roughly the top 2 or 3 per module, around 16 to 24 genes. A good shortlist gene has a high kME, a functional summary that mentions cancer or a relevant process, and a non-trivial altered percentage in the cancer columns. Only those shortlisted genes go through the web tools below.

Tool	Usage
UALCAN or GEPIA2	Boxplots of gene expression in tumor versus normal breast tissue, and expression split by PAM50 subtype (Luminal A, Luminal B, HER2 enriched, Basal or TNBC). This confirms the gene is actually altered in tumors and tells which subtype a module points to.
KM Plotter	Kaplan-Meier survival curves. It splits patients into high and low expression of the gene and tests whether that split predicts overall or relapse free survival. A significant split (low log-rank p-value) means the gene is a candidate prognostic marker.
STRING-db	A protein-protein interaction network.
cBioPortal OncoPrint	The same cohort the script queried, viewed interactively. The OncoPrint is a barcode style chart of which patients carry mutations, amplifications, or deletions in our shortlisted genes, and whether those events co-occur with known drivers such as TP53 or PIK3CA.